

Relational memory in multi-agent simulations

Modeling human-like memory
for more believable simulations

Andre Atkins, Ben Sadorra, and Ryan Shechtman
CMSC473/673

The problem: human-like remembering

A familiar place can bring back an old experience.

The pattern we want to model

An experience can be hard to recall.
A specific context can bring it back.
Other experiences continue to fade.

The gap in a simple retrieval rule

Text similarity finds matching content.
Recency favors recent experiences.
Personal associations need their own history.

Our goal is believable remembering and forgetting in a social simulation.

Example: remembering where Mia helped

Earlier Mia gives the agent food beside a blue pinwheel.
She says, “I’m usually here.”

Much later The hungry agent notices the same pinwheel.
Mia and food are outside its current view.

The reminder The pinwheel brings back that encounter,
giving the agent a reason to look for Mia there.

Expected behavior: approach the location and seek Mia’s help.

Thesis and testable hypotheses

Associative memory should give agents more appropriate context for their decisions, making their behavior more believable and consistent with their own experiences.

H1 Associative recovery

Associations will help agents retrieve past experiences relevant to their present situation, including experiences that semantic similarity and time decay alone would miss.

H2 Lasting accessibility

Recalling experiences through their associations will help preserve relevant context across longer gaps.

Models and their roles

Role	Planned model	Rationale
Direct similarity	SBERT all-MiniLM-L6-v2	Sparser similarity scores in the cited setup. Small enough to run locally.
Actions and messages	GPT-5 Nano	API generation for repeated matched simulation runs.
Lower API-cost option	A locally available Qwen model	Local generation for larger collections when the hardware permits.

SBERT scores the cue against memory text.

The generation model receives memories before acting.

The memory's own forgetting curve

Direct relevance r_i

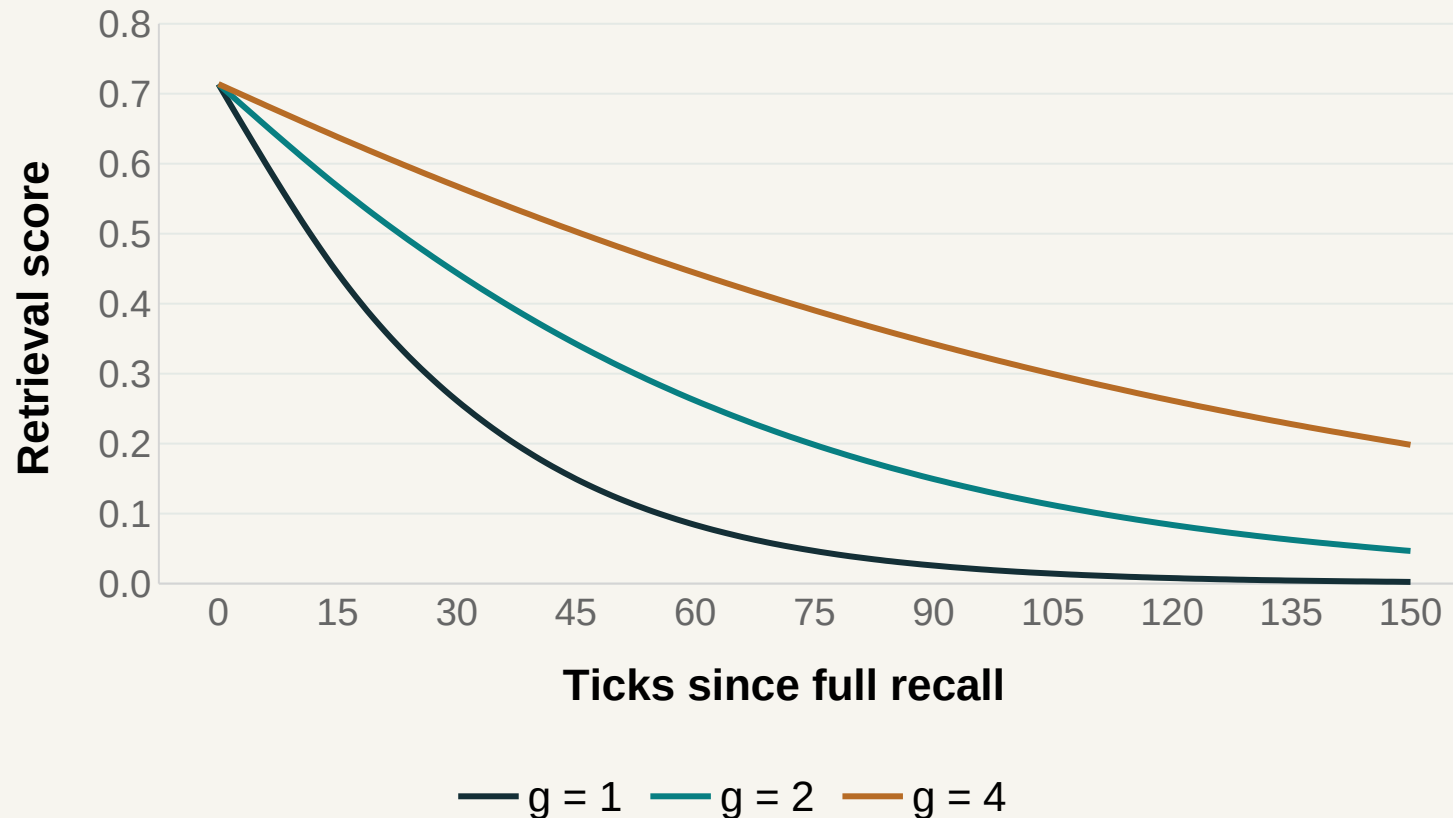
Cue-to-memory
embedding similarity

Elapsed time

Time since full recall

Memory strength g_i

Each memory has its own g .
A higher g slows forgetting.



Calculated Hou curves at relevance .60. These are model scores, not measured human recall rates.

A separate pathway for associations

Direct-memory contribution

$$M_i = r_i \exp(-t_i / (\tau_m g_i))$$

r_i cue-to-memory cosine similarity

g_i this memory's strength

τ_m tick-to-memory time scale

$$x_i = \max(M_i, G_i)$$

One uncrowded connection

$$G_i = \beta c w \exp(-\Delta / h)$$

c current cue activation

w connection weight

h connection retention time

$$p_i = (1 - \exp(-x_i)) / (1 - \exp(-1))$$

Each route ages separately. Max takes the stronger contribution. β limits the graph contribution.

When the pinwheel can bring it back

Ticks 0–32

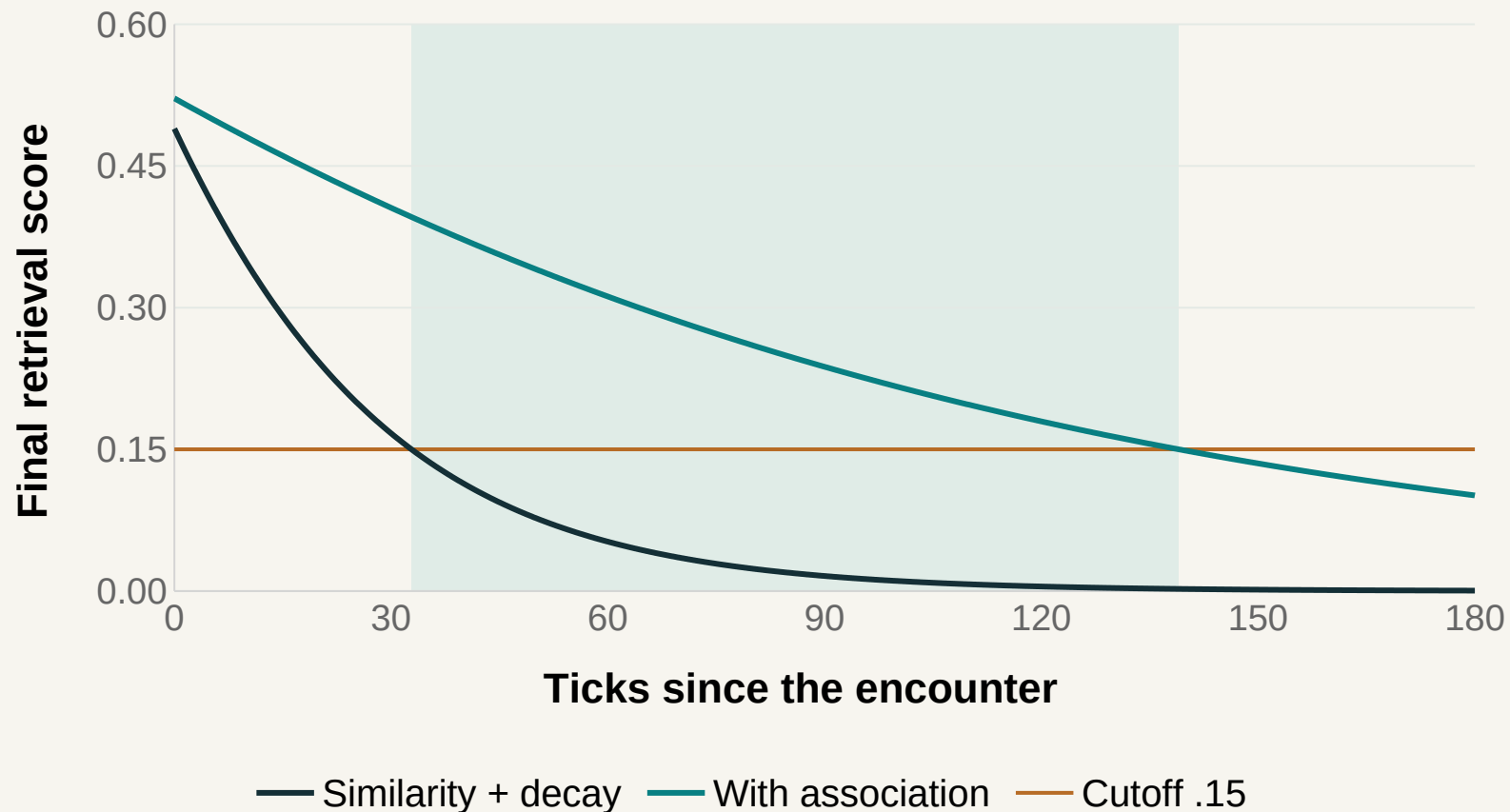
Either method can retrieve the encounter.

Ticks 33–139

Only the association keeps it above .15.

From tick 140

Both fall below the cutoff.



One fixed episode and visible cue. No intervening recall, new connections or competing memories.

Which memories enter the prompt?

- 1 Score the memories** Compare direct and associative contributions before changing any strengths.
- 2 Fill the memory slots** Take up to 3 highest-scoring memories above .15 that fit within 600 tokens.
- 3 Apply recall updates** Rehearse each included memory. Update its strongest graph route only if it beats direct access.

“Selected” means its text enters the agent’s prompt.

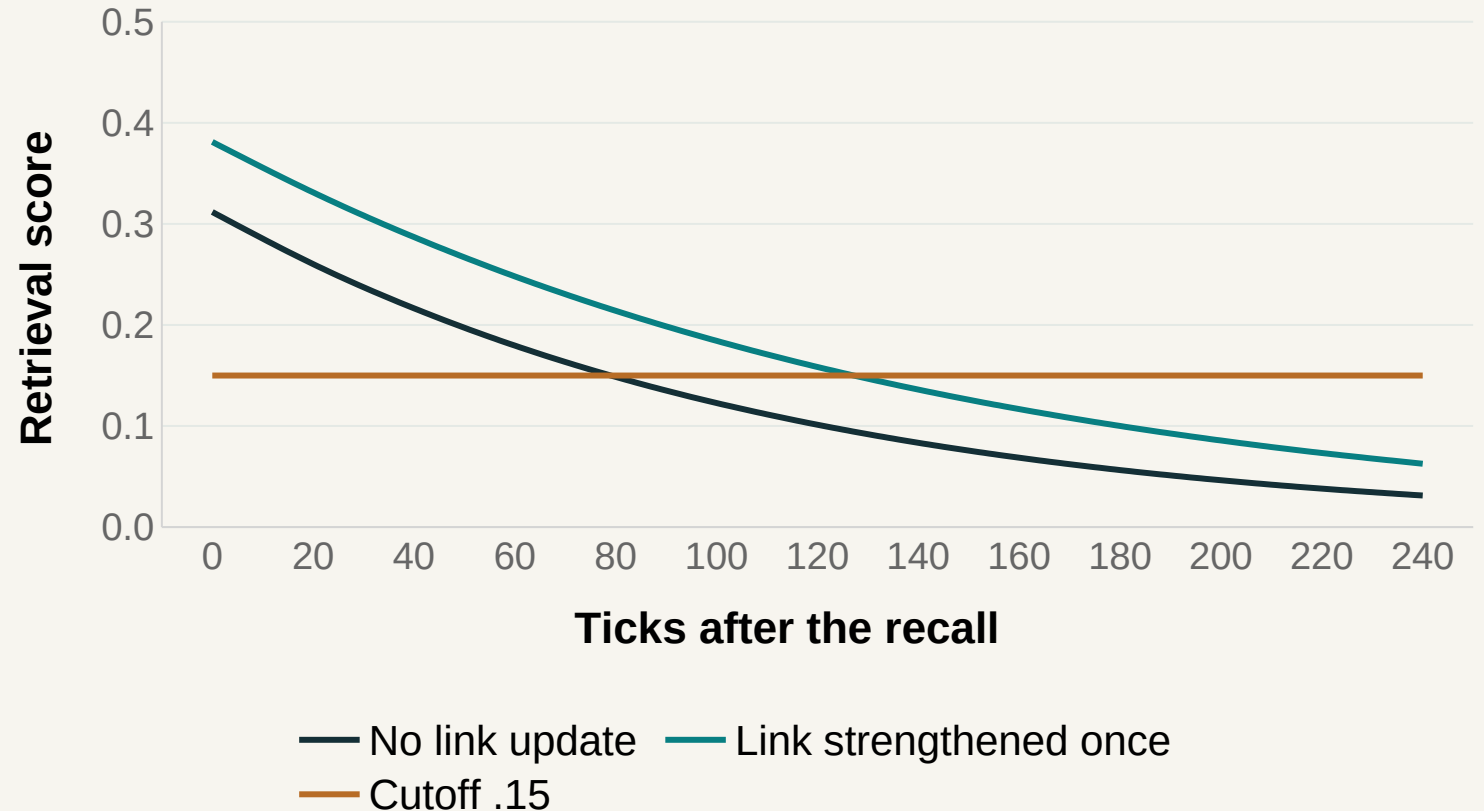
Connection rehearsal and later access

Both recall the same episode once

Only one also strengthens the connection.

The score clears the cutoff longer.

Both links still fade.



Fixed graph and cue. Separate probes, with no new episodes or intervening recalls.

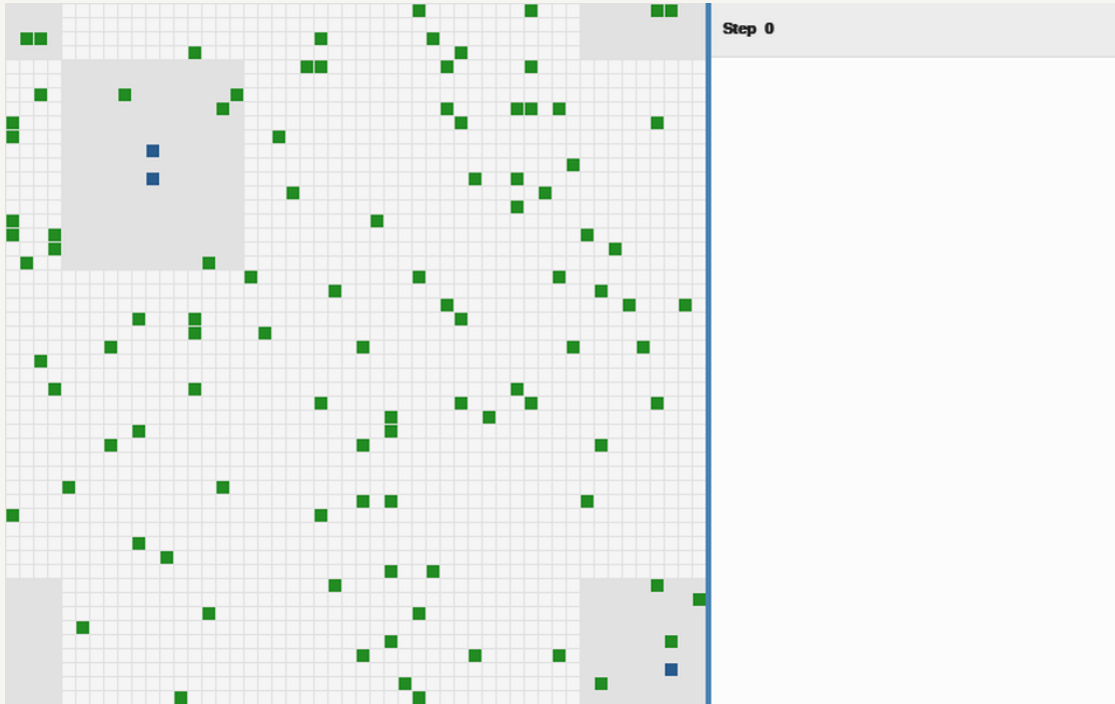
Experiment conditions

Condition	What it tests
Semantic similarity and ordinary decay	Baseline remembering and forgetting
Associative retrieval, link learning off	H1: relevant experiences recovered through surviving connections
Associative retrieval, link learning on	H2: useful context preserved after a connection helps recall an experience

We will vary the time gap and the number of competing memories.

Additional checks: goal-aware queries, slower ordinary forgetting, wrong cues and crowded places.

TerraLingua and the memory audit UI



The agent's input

Local observations and messages

State, plan and recalled experiences

Planned audit trail

Direct and associative contributions

Winning route and edge age

Memories included and left out

Changes to memory and link strength

Trace each retrieval and state update, then inspect the exact prompt and resulting action.

Evaluation: memory patterns and believability

Question	Evidence we will collect
H1: does context recover relevant old episodes?	Relevant recall, false reminders and displacement
H2: does rehearsal preserve useful access?	Useful and distracting reminders after longer gaps
Do decisions fit the recalled context?	Prompt and action traces, plus blind ratings
Does the simulation feel more believable?	Blind ratings of remembering and behavior

Prompt inclusion, appropriate decisions and believability are separate outcomes.

Proposed implementation and experiments

- 1 Reference scoring and traces**

Separate the memory and edge clocks. Verify the worked examples.
- 2 Controlled comparisons**

Vary delays and distractions. Test retrieval and connection learning.
- 3 Simulation and human review**

Integrate the audit UI and evaluate believable remembering.

Initial scope: independent associative access and connection learning.

Full retrieval notation

q: current cue **m_i:** stored episode **E:** SBERT embedding

$$r_i = \max(0, \cos(E(q), E(m_i)))$$

$$M_i = r_i \exp(-(t - t_{i,\text{last}}) / (\tau_m g_i))$$

$$L_e = w_e \exp(-(t - t_{e,\text{last}}) / h_e)$$

$$G_i = \beta \max \text{ over routes } [\text{source cue} \times \text{edge transmissions} \times \text{crowd penalty}]$$

$$x_i = \max(M_i, G_i) \quad p_i = (1 - \exp(-x_i)) / (1 - \exp(-1))$$

The prompt receives at most k memories above the cutoff that fit within the token budget.

What changes after a retrieval

State	Update	When
Memory strength g and clock	Increase g by the Hou spacing rule. Reset its last-full-recall time.	The memory enters the prompt.
Connection strength w	Start from current transmission z . Set $w = z + \eta(1 - z)$.	Its memory enters the prompt and this graph route beats direct access.
Connection duration h and clock	Raise h to a finite cap. Record this reinforcement time.	Once per qualifying edge per retrieval event.
Memories left out	No recall update. Their connections continue aging.	Cutoff, slot limit or token limit excludes the memory.

A preview or merely seeing the cue changes no existing memory or connection strength.

Graph sources and allowed connections

Connection	Meaning	Initial scope
Entity to episode	A current entity matches an episode's entity tag.	Main condition: explicitly observed entity cues.
Same tick	Two separately stored events share a creation tick.	Optional. The writer normally stores one episode per turn.
Adjacent episodes	One stored episode follows another.	Optional. Immediately previous or next stored episode.
Memory through one entity hub	A direct source cues another episode through two mentions edges.	Optional. Fixed seed budget and no recursive spreading.

Example: receiving food beside a pinwheel links the encounter to that object. The object did not cause the help.

Time and the experimental controls

Test	What varies	Same at each time gap
H1: associative recovery	Time gap, competing memories and graph access on or off	Identical memory state, cue and prompt budget.
H2: lasting accessibility	Time since the same recall event and connection learning on or off	Equal ordinary recalls. Only one copy updates its link.
Believability in the simulation	Memory condition across repeated simulation runs	Same generation settings and prompt budgets.

Time still causes decay. We vary the gap and compare at each gap.

Expected limits and failure cases

Failure case	What we expect or need to measure
No active associated cue	No support through that cue. Other routes may still work.
Several episodes share a cue	Graph scores can override differences in text relevance.
A cue accumulates many episodes	Crowding can push even a surviving link below the cutoff.
Repeated false reminder	An irrelevant association can strengthen and recur.
Aging of stored content	Access changes. Stored text stays intact.

Cue-triggered retrieval is a modeling hypothesis. Clinical trauma and emotion require additional mechanisms.

Questions we should be ready to answer

Question	Answer
Does the example prove vector search fails?	Similarity is positive. Ordinary decay puts the old encounter below the cutoff.
Is graph structure necessary here?	A graph or weighted lookup can store the same cue-to-episode associations.
Does reinforcement prove better memory?	It raises support by design. Useful recall and believability still need evaluation.
What does selected mean?	Its text enters the prompt after the cutoff and the memory and token limits.
What can max lose?	When graph support dominates, differences in direct text relevance can stop affecting rank.

Sources

Source	Role in the project
Hou, Tamoto & Miyashita (2024)	Direct-memory decay and rehearsal
Honda et al., HAI 2025 proceedings	Embedding choice and memory-model context
Anderson & Reder, 1999	Associative competition and the fan effect
Paolo et al., TerraLingua, 2026	Persistent multi-agent simulation environment
APA Dictionary of Psychology	Associative-memory motivation
Kuroki et al., Shachi	Agent-believability evaluation context
Huang et al., EmotionBench	Secondary behavioral-evaluation context
Course proposal and design brief	Project goal, proposed equations and worked examples

Full references appear in the speaker notes. The independent-pathway equations are our proposal.